Q1. Is this a regression or classification problem? Give two reasons based on what the target represents and how its values behave.

- This is a regression problem.
- Reason 1: If we consider target value behavior, it spans a continuous numeric range with many distinct, non-discrete values.
- Reason 2: If we consider target representation, the target variable represents a continuous, quantitative measure of disease progression one year after baseline instead of specific category labels.

Q2. Why does a numeric target not automatically imply a regression problem? Give one example of a numeric class label.

- Numeric values can be used to encode discrete, non-continuous categories where arithmetic operations do not make sense.
- Encoding customer survey ratings or binary classes as 0 and 1 (ZIP codes / categorical IDs as numbers).

Q3. Describe the target distribution, including its spread, overall shape and any noticeable extreme values.

- Target values range from a minimum of 25 to a maximum of 346, with a mean around 152.13 and a standard deviation around 77.09.
- The distribution is roughly unimodal and slightly right-skewed, with a higher concentration of values in the lower-to-middle range (75–180) and a tapering tail toward upper extreme values near 300+.

Q4. From the feature-target plots, identify two features with relatively clear linear trends and one feature with a weak relationship. State whether the observed trends are positive or negative.

- Clear linear trends are **bmi** and **s5**
- Weak relationship is **s2**

Q5. Do any feature-target plots show noticeable curvature, large scatter or unusual observations? Identify at least one example and explain why it matters when considering linear regression.

- **s2** and **age** show large scatter with wide vertical variance across target values.

- If data points are scattered all over the place without a clear trend, a straight line simply can't fit them well. This means the model will struggle to make reliable predictions and will not explain much of the variance in disease progression.

Q6. List the three features with the largest absolute correlation with the target. Also identify one strongly correlated pair of input features and state why predictor-predictor correlation can matter in multiple linear regression.

- Correlated Features: **bmi**, **s5**, **bp**
- Strongly Correlated Feature Pair: **s1** and **s2**
- When two predictors move together, standard linear regression struggles to separate their individual effects on the outcome. This inflates the uncertainty around the model's coefficients, making feature importances unstable and hard to trust, even if the overall prediction accuracy of the model remains good.

Q7. Before fitting the prediction model, is multiple linear regression a reasonable model to try? Justify your answer using at least three observations from Parts A and B.

- Continuous Target: The target is continuous and quantitative with a smooth distribution.
- Linear Feature Trends: **bmi**, **s5**, **bp**
- Strong Correlations: Multiple features show moderate-to-strong correlation values with the target without severe non-linear patterns.

Q8. Report the baseline RMSE and R-squared. What does this baseline represent, and why is it useful?

- Baseline RMSE: ~55.61
- Baseline R^2: 0.00
- The reason is it represents a naive predictor that ignores all input features and outputs the overall training target mean. It serves as a benchmark floor, any useful model must perform significantly better than this baseline.

Q9. Report the linear-regression RMSE and R-squared. Does it outperform the mean baseline? Support your answer using the metric values.

- Linear Regression RMSE: ~53.48
- Linear Regression R^2: ~0.452
- Yes, linear regression significantly outperforms the baseline. It reduces test RMSE from ~55.61 down to ~53.48 and improves R^2 from 0.00 to ~0.45

Q10. Inspect the actual-versus-predicted and residual plots. Do you observe any systematic pattern that suggests the linear model is missing structure? Explain briefly.

- Points scatter reasonably well along the y = x reference line, though variance increases slightly for higher target values.

- The residual vs predicted plot shows a relatively random scatter around zero without strong U-shapes or severe funneling.

- So there is no major missing non-linear structure, though slight fan-out at higher target ranges suggests potential minor unmodeled variance.

Q11. Choose one unseen test observation from Part E. Report its predicted target, actual target and absolute error. In one sentence, explain what model.predict(...) is doing in this step.

- Actual Target: 219.00
- Predicted Target: 139.55
- Absolute Error: 79.45

- The model.predict(...) function takes the test feature values for that patient, multiplies each feature by its corresponding learned model weight, and adds the intercept to calculate a final predicted progression score.